

Retail Demand Forecasting & Inventory Optimization

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Project Overview

Objective: Forecast retail demand and optimize inventory using Python, Streamlit, Prophet, LightGBM and Google BigQuery.

Dataset & ETL

Retail sales dataset cleaned, transformed and prepared for analysis. ETL includes loading, cleaning, validation and storage.

EDA

Performed descriptive statistics, correlation analysis, demand distribution, monthly trends, category analysis and promotion analysis.

Business Insights

Forecasting reduces stockouts, improves inventory planning, supports procurement and lowers warehousing costs.

Machine Learning

Prophet captures trend and seasonality for time-series forecasting. LightGBM predicts demand using multiple business features.

Dashboard

Power BI and Streamlit dashboards provide KPI cards, filters, charts and downloadable reports for decision-making.

Conclusion

The solution enables proactive inventory management and data-driven retail decisions.